

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

By

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Project Tile – AI Agent for Smart Farming Advice

Domain of Project – Agriculture

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AI-Powered Smart Farming Agent

Intelligent · Personalized · Real-time Farming Advisory

KEY FEATURES



Agricultural Knowledge (RAG)

Retrieves crop best practices, government advisories and farming guidelines.



Weather & Irrigation Insights

Uses weather forecasts and soil data to optimize irrigation and farm operations.



Pest & Disease Detection

Analyzes crop images to identify diseases, pests and nutrient deficiencies.



Personalized Farm Advisory

Recommends crops, seeds, fertilizers, irrigation schedules and harvesting timelines.



Market Intelligence

Provides price forecasts, demand trends and cost-optimization suggestions.

MULTI-AGENT ARCHITECTURE



INPUT DATA SOURCES



Weather Data

Real-time forecasts, temperature, rainfall, humidity, wind.



Soil & Farm Data

Soil health, moisture, nutrients, pH, land characteristics.



Crop Images

Images of leaves, stems, fruits for pest/disease detection.



Market Data

Prices, demand trends, supply, mandi/market updates.



Knowledge Bases

Agricultural research, government portals, best practices.

TECHNOLOGY STACK



LangFlow
Workflow & Agent Orchestration

watsonx.ai

Model Deployment, Embeddings & Governance



Granite Models
Reasoning, Summarization, Recommendations & Advisory



Vector Database
RAG for Accurate Knowledge Retrieval



Streamlit / React
Dashboard & Insights Visualization

HOW IT WORKS



Farmer Input
Query, Image, Location, Crop & Land Details



Multi-Agent Processing
Agents collaborate and analyze data



AI Reasoning & RAG
Retrieves knowledge and generates recommendations



Actionable Advice
Personalized, real-time and context-aware suggestions

EXPECTED IMPACT

- ✓ Higher crop productivity and yield
- ✓ Efficient use of water, fertilizers and resources
- ✓ Early detection of pests and diseases
- ✓ Data-driven decisions for better income and profitability
- ✓ Sustainable and climate-resilient farming



Empowering Farmers with Intelligent Insights for a Better Tomorrow

AI Agent for Smart Farming Advice Agent

Problem Statement :

Farmers often struggle with low crop yield, unpredictable weather, pest infestations, and limited access to timely expert advice. Traditional farming advisory services are fragmented, slow, and not personalized to the farmer's specific land, crop, or climate conditions. With the rise of climate change, data-driven smart farming solutions are needed to empower farmers with actionable insights.

The Objective :

Develop an AI-powered Smart Farming Agent that provides personalized, real-time, and sustainable agricultural advice. The solution should:

1. **Agricultural Knowledge Retrieval (RAG)** – Use Retrieval-Augmented Generation to fetch best practices, crop-specific guidelines, and government advisory updates.
2. **Weather & Soil Insights** – Integrate real-time weather forecasts, soil health data, and irrigation needs to optimize farm planning.
3. **Pest & Disease Detection** – Enable multimodal support where farmers upload crop images for early pest/disease identification and get instant recommendations.
4. **Personalized Advisory Agent** – Suggest seeds, fertilizers, irrigation schedules, and harvesting timelines based on farmer profile and land conditions.
5. **Market & Cost Optimization** – Forecast demand, suggest market prices, and recommend cost-efficient practices for better income.
6. **Multi-Agent Collaboration** – Crop Advisory Agent o Weather & Irrigation Agent o Pest/Disease Detection Agent o Market Insights Agent

Technology Used

Tech Stack (Mandatory)

Use **any one** of the following approaches:

- LangFlow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with LangChain + RAG + IBM Granite Models

I used LangFlow + IBM watsonx.ai + Granite Models

Proposed Solution -

Proposed Solution: AI-Powered Smart Farming Agent

An intelligent multi-agent farming assistant built using **LangFlow**, **IBM watsonx.ai**, and **Granite Models** to deliver personalized, real-time agricultural guidance.

Key Features

Agricultural Knowledge (RAG): Retrieves crop best practices, government advisories, and farming guidelines from trusted sources.

Weather & Irrigation Insights: Uses weather forecasts and soil data to optimize irrigation and farm operations.

Pest & Disease Detection: Analyzes crop images to identify diseases, pests, and nutrient deficiencies with treatment recommendations.

Personalized Advisory: Recommends crops, seeds, fertilizers, irrigation schedules, and harvesting timelines based on land and farmer profiles.

Market Intelligence: Provides price forecasts, demand trends, and cost-optimization suggestions.

Multi-Agent Architecture

- Crop Advisory Agent
- Weather & Irrigation Agent
- Pest & Disease Detection Agent
- Market Insights Agent

Technology Stack

LangFlow • IBM watsonx.ai • Granite Models • RAG & Vector Database • Streamlit/React Dashboard

Impact

Enables data-driven farming through timely insights, improved crop productivity, efficient resource utilization, reduced risks, and enhanced farmer profitability.

Langflow component Used -

- **Chat Input** – Captures farmer queries and requests.
- **Prompt Template** – Structures prompts for advisory generation.
- **Granite LLM (watsonx.ai)** – Provides reasoning and personalized recommendations.
- **RAG Pipeline** – Retrieves agricultural knowledge, crop guidelines, and government advisories.
- **Vector Store** – Stores and retrieves embedded farming documents efficiently.
- **Retriever Component** – Fetches relevant information from the knowledge base.
- **Multi-Agent Flow** – Coordinates Crop Advisory, Weather & Irrigation, Pest Detection, and Market Insights agents.
- **Image Input Component** – Accepts crop images for pest and disease analysis.
- **API/Tool Components** – Integrates weather, soil, and market data services.
- **Output & Dashboard Components** – Delivers recommendations and visual insights to farmers.

Langflow workflow -

The screenshot displays the IBM Watson Studio interface for a project named "Agri-Doctor". The top navigation bar shows the project name and a "Starter Project" label. The left sidebar contains a search bar and a list of components: Input & Output, Data Sources, Models & Agents, LLM Operations, Files & Knowledge, Processing, Flow Control, Utilities, and a "Discover more components" button. The main workspace shows a workflow diagram with the following components and connections:

- Chat In...** (27ms) connects to **IBM watsonx.ai** (1.4s).
- IBM watsonx.ai** connects to **officer 1** (2.5K | 3.9s).
- officer 1** connects to **Officer 2** (2.7K | 2.7s).
- Officer 2** connects to **Chat O...** (24ms).

The **IBM watsonx.ai** component configuration includes:

- Language Model:** Selectable dropdown.
- Input:** Text input field.
- System Message:** Text input field.
- watsonx API Endpoint:** Text input field with a dropdown arrow.
- watsonx Project_ID:** Text input field with a dropdown arrow.
- watsonx Space_ID:** Text input field with a dropdown arrow.
- Watsonx API Key:** Text input field with a dropdown arrow.
- Model Name:** Text input field.
- Language Model:** Selectable dropdown.

The **officer 1** and **Officer 2** components have similar configurations for **Language Model**, **Agent Instructions**, **Tools**, and **Input**.

The bottom status bar shows a zoom level of 56% and various utility icons.

The screenshot displays the IBM Watson Studio interface for a project named "Starter Project / Agri-Doctor". The top navigation bar includes a search bar, a "Playground" button, and a "Share" dropdown. The left sidebar contains a "Components" panel with categories like "Input & Output", "Data Sources", "Models & Agents", "LLM Operations", "Files & Knowledge", "Processing", "Flow Control", "Utilities", and "Discover more components". The main workspace shows a flow diagram with three components: "Chat In...", "IBM watsonx.ai", and "Officer 2". The "IBM watsonx.ai" component is configured with the following settings:

- Language Model:** ibm/granite-8b-code-instruct
- Agent Instructions:** officer 1: Plant Pathology Expert
- Tools:** (empty)
- Input:** Receiving input
- System Message:** Type something...
- watsonx API Endpoint:** https://au-syd.ml.cloud.ibm.com
- watsonx Project_ID:** 96978457-5977-47fb-b2b1-42e0...
- watsonx Space_ID:** Type something...
- Watsonx API Key:** (masked)

The "Officer 2" component is also configured with similar settings, including the "Agent Instructions" set to "officer 2: Agricultural Extension Officer". A green notification box at the bottom center of the workspace states "Flow built successfully". The bottom status bar shows a progress indicator at 56% and various utility icons.

Flow built successfully

8.5s

Langflow Text Response-

←

→

↺

📄 localhost:7860/flow/03b89e7e-21c0-4125-b30f-0e12516838fd

☆

⬇

🔒 Enter passphrase

⋮

🌀

Starter Project / 📄 Agri-Doctor

👤 150k

💬 25k

🔔

👤

Sessions

+

Default Session

⋮

☐ Select All

⋮

☐ New Session 0

⋮

☐ New Session 1

⋮

☒ New Session 2

⋮

New Session 2

✕

👤 User

The older leaves on my maize (corn) crop are developing a distinct V-shaped yellowing pattern starting from the tips and moving down the center vein, while the rest of the leaf stays green. What is missing from my soil, and how can I fix this quickly and sustainably?

🌀 AI

✓ Finished in 1.2K | 8.3s

Called tool GET CURRENT DATE

7ms

Called tool ACCESSING GET CURRENT DATE

10ms

The missing nutrient in your soil is most likely nitrogen. To fix this quickly and sustainably, you can add a nitrogen-rich fertilizer to your soil. You can also consider using cover crops or compost to help improve the soil's nitrogen content. Additionally, you can adjust your irrigation schedule to ensure that

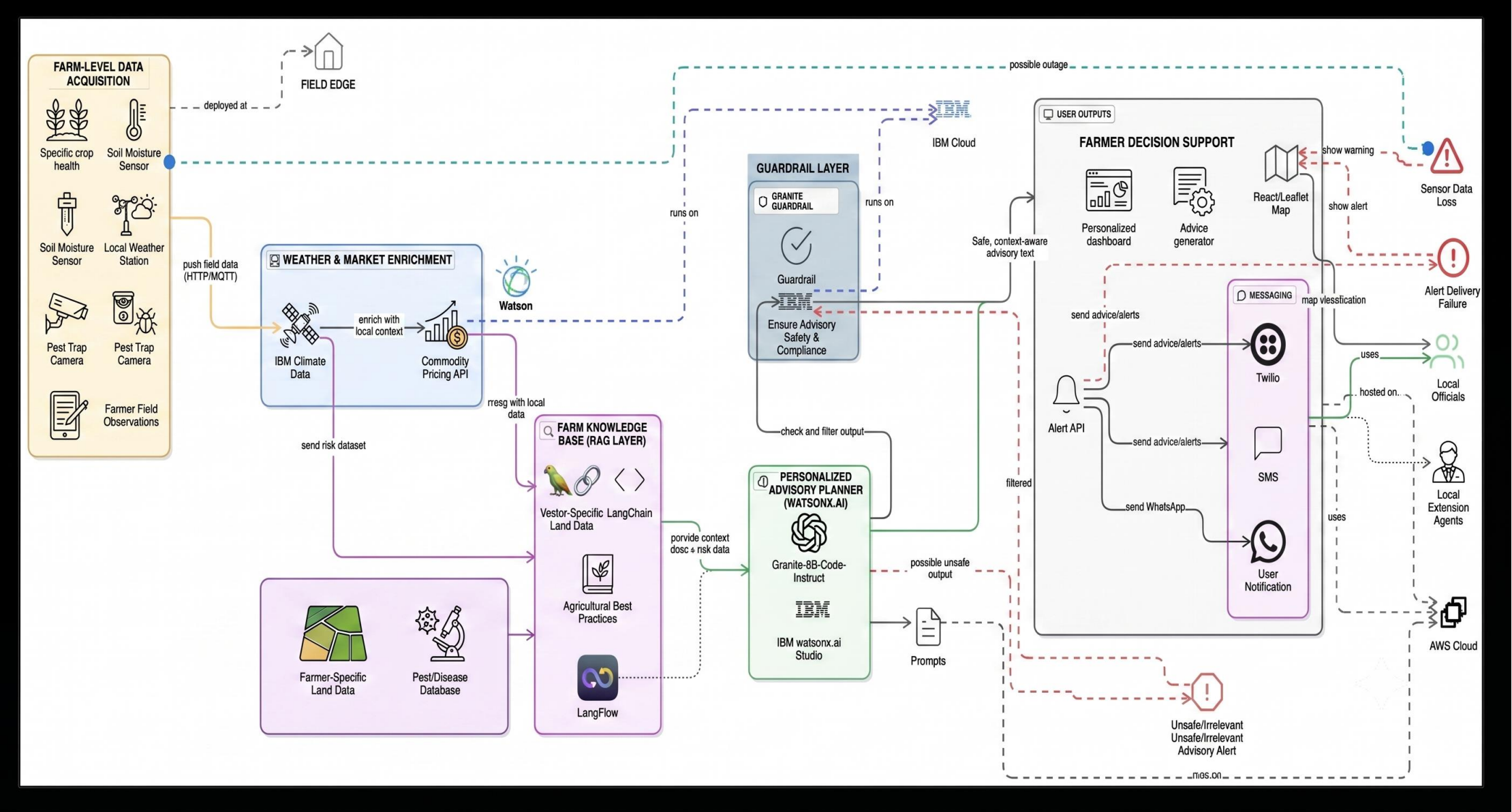
Send a message...

📄

🗣

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Architecture blueprint



Role of Agentic AI in the solution

Agentic AI serves as the decision-making and coordination layer of the Smart Farming Agent. Instead of relying on a single AI model, multiple specialized agents collaborate autonomously to analyze data, retrieve knowledge, and generate actionable recommendations.

Crop Advisory Agent provides personalized recommendations on crops, seeds, fertilizers, and harvesting schedules.

Weather & Irrigation Agent analyzes real-time weather and soil conditions to optimize irrigation and farm planning.

Pest & Disease Detection Agent identifies crop health issues from uploaded images and suggests corrective actions.

Market Insights Agent evaluates market trends, demand forecasts, and pricing information to improve farmer profitability.

These agents communicate through LangFlow workflows, share insights, and collectively deliver accurate, context-aware, and personalized farming advice. This Agentic AI approach enables real-time decision support, improves resource utilization, reduces farming risks, and enhances crop productivity and income generation.



Technology Used

Core Technology Stack & Multi-Agent Workflow

1. Workflow Orchestration Platform

- **LangFlow Engine:** The system is built and run as a LangFlow graph graph (running locally at `localhost:7860`), allowing visual construction of multi-agent workflows.

2. Base Language Model (LLM) Configuration

- **Provider Integration Component:** `IBM watsonx.ai` node.
- **Model Selected:** `ibm/granite-8b-code-instruct` .
- **API Configuration Endpoint:** Deployed using the Asia-Pacific regional instance (`https://au-syd.ml.cloud.ibm.com`).
- **Resource Scope:** Validated through a dedicated `watsonx Project_ID` (`96978457-5977-47fb-b2b1-[REDACTED]`).

Technology Used

Core Technology Stack & Multi-Agent Workflow

3. Sequential Multi-Agent Architecture

The system dynamically routes data sequentially through two specialized agent components:

- **Officer 1: Plant Pathology Expert**
 - **Role:** Analyzes raw queries or crop conditions to identify plant diseases, pest infestations, or nutrient deficiencies, outlining scientific control treatments (Organic, Chemical, and Integrated Pest Management).
 - **Input Connectivity:** Receives the original user request directly from the `Chat Input` component.
 - **LLM Engine:** Connected directly to the centralized `IBM watsonx.ai` model output node.

Technology Used

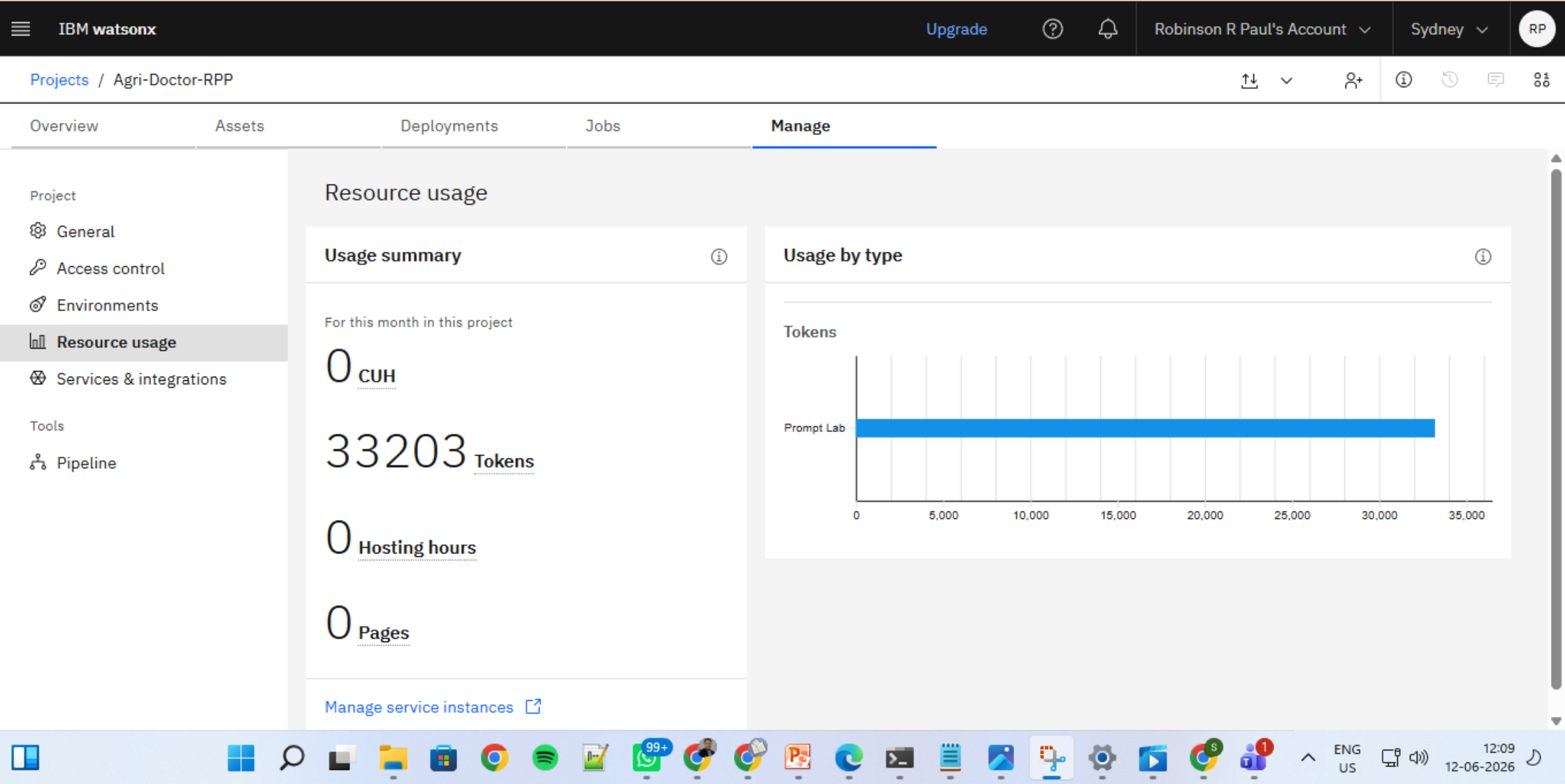
Core Technology Stack & Multi-Agent Workflow

- **Officer 2: Agricultural Extension Officer**
 - **Role:** Translates the scientific, complex recommendations generated by Officer 1 into practical, non-technical field actions, localized step-by-step guidance, and immediate 24–48 hour action plans for the farmer.
 - **Input Connectivity:** Chains directly into the output (`Response`) of Officer 1, converting the technical findings into farmer-friendly deliverables.
 - **LLM Engine:** Linked directly to the same centralized `IBM watsonx.ai` model output node.

4. Output Delivery

- **Chat Output Component:** Captures the final simplified text generated by Officer 2 to render the clear, actionable field execution plan directly inside the user interface.

Token Usage



Windows taskbar with various application icons (File Explorer, Chrome, Spotify, etc.) and system tray showing ENG US, 12:09, and 12-06-2026.

Novelty and Uniqueness

The proposed Nutrition Agent stands out by combining **Agentic AI, RAG, and personalization** into a single intelligent system.

Multi-Agent Intelligence – Instead of a single chatbot, multiple specialized agents collaborate to deliver accurate and context-aware nutrition guidance.

RAG-Based Accuracy – Integrates real-time data retrieval with AI generation, reducing misinformation and ensuring trustworthy recommendations.

Hyper-Personalization – Tailors diet plans based on health conditions, cultural preferences, and fitness goals.

Real-Time Feedback Loop – Continuously adapts recommendations based on daily food logs and user behavior.

Multi-Modal Input – Supports text, voice, and image-based meal tracking for better usability.

Preventive Healthcare Focus – Goes beyond diet planning to provide disease-specific nutrition advice.

Scalable & Enterprise-Ready – Built on **IBM watsonx.ai** with **IBM Granite models**, ensuring reliability and governance.

This makes the solution a **smart, adaptive, and proactive digital nutritionist**, not just a static diet recommendation tool.

Git Hub Link

<https://github.com/ursbestfriend/IBMskilbuild-Langflow-Agri-Doctor>

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Future Scope

1. IoT Integration:

Ingests live soil data and microclimate telemetry to deliver highly dynamic, precise irrigation and fertilizer plans.

2. Voice & Multilingual Support:

Implements natural voice-enabled interfaces in regional vernacular dialects, ensuring inclusive and seamless field accessibility.

3. Multimodal Vision Processing:

Leverages smartphone leaf photo uploads for instantaneous visual pest and disease identification.

4. Dynamic RAG Systems:

Connects to live agricultural databases via Retrieval-Augmented Generation to ensure verified, hallucination-free guidelines.

Thank You!

Thank you for your time and interest.